

Model Card & Project Performance Report

Customer Feedback Intelligence & Root-Cause Triage Engine

1. Executive Summary & Objective

E-commerce platforms receive high volumes of unstructured customer feedback. This system implements a two-tier NLP pipeline: **Tier 1** predicts polarity sentiment (Positive vs Negative) via tuned LinearSVC, and **Tier 2** flags negative feedback into 5 operational defect categories.

2. Dataset Specifications

- **Domain:** Customer Reviews (Electronics)
- **Sample Size:** 8,000 Balanced Records
- **Target Task:** Binary Polarity Classification

3. NLP Preprocessing

- **Normalization:** Lowercasing, whitespace cleanup
- **Features:** TF-IDF (1-3 n-grams, Sublinear TF)
- **Vocabulary Cap:** 12,000 max features

4. Model Benchmarks & Validation

Model Architecture	Representation	Accuracy	Macro F1	Status
Multinomial Naive Bayes	TF-IDF (1-2 n-grams)	82.19%	0.82	Baseline
LinearSVC (Tuned, C=0.1)	TF-IDF (1-3 n-grams, Sublinear)	86.69%	0.87	Champion Pick

5. Root-Cause Operational Buckets

Hardware & Battery: Battery drain, switches, display defects

Shipping & Logistics: Courier delays, arrival status, damages

Support & Warranty: Return issues, warranty bottlenecks

Build Quality: Broken components, fragile casing, flaws

Pricing & Value: Price-to-value gap, high cost complaints

General Defect: Non-specific performance remarks

Tech Stack: Python, Scikit-learn, Pandas, NLTK, Streamlit

Saved Artifacts: best_sentiment_model.pkl, best_tfidf_vectorizer.pkl